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**Is Good always Right? The The
Instability of Space -Valence Body-
Specific Associations**

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of the requirements for the degree of M. Sc. in Cognitive Science

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Violeta Valkova.....



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Abstract

According to the body-specificity hypothesis ([Casasanto, 2009](#)) people who interact with the environment differently (e.g., right- and left-handers), should also construct different mental representations even of abstract concepts such as “good” and “bad” depending on their handedness. [Casasanto \(2009\)](#) showed that people associate positive concepts with the space around their dominant hand, and negative concepts with the space around their subordinate hand. First, previous findings were replicated and extended, establishing a relationship between handedness and mental representation of abstract concepts. In a task which required categorization of positive, and negative nouns participants showed the expected RT-profiles: right- handers categorized positive concepts faster with their right hand than with the left one, and negative concepts faster with their left hand than with the right one. Left-handers showed the opposite pattern. The third examined group, ambidextrous, showed no differences in the categorizing speed for positive and negative words. Second, the question was whether short-term changes in motor fluency could reverse the association between the space and abstract concepts to the extent visible in the online valence categorization task. The hand motor fluency of our right-handed participants was reduced applying a Domino task ([Casasanto & Chrysikou, 2011](#)). Then the same participants were run through the RT valence categorization task. The results showed a non-significant interaction with the same pattern obtained by left-handers discussed above. To conclude, the results showed high level of instability of our mental representations of abstract concepts, and their dependency on currently experienced motor fluency.

Introduction

For decades embodiment (e.g., [Barsalow, 1999](#)) attempts to take precedence over the “computer metaphor” of the brain (e.g., [Marr, 1982](#)). A growing body of empirical evidence opposes the idea that mental representation are just symbols on which computations are performed emphasizing the important role of the interaction of the own body with the environment ([Fischer & Coello, 2015](#)). Most of the abstract concepts are a great challenge to the embodied condition, because one cannot have any bodily experience with them. Therefore they should be represented in a different way than the concrete concepts. A solution is provided by the conceptual metaphor theory ([Lakoff & Johnson, 1980](#)) according to which abstract concepts are represented in terms of more concrete concepts. That explanation is further extended by [Casasanto, 2009](#) who suggests that the concepts of the e.g. emotional *valence* are represented in terms of *space*, and that, those representations differ depending on our bodies. [Casasanto, 2009](#) proposed the body-specificity hypothesis according to which people who interact with the environment differently form also different mental representations. Different studies showed that people conceptualize *good* with the help of the space around their dominant hand and *bad* – with the help of the space around their subordinate hand ([Casasanto, 2011, 2014](#) for review). Another line of research showed that the conceptualization of a concept might depend on the currently experienced motor fluency in the interaction with the environment. The current research tries to extend the dependence results to the processing stability of the associations between space and valence. Chapter 1 and 2 discuss the theories of conceptual knowledge representation and in particular the conceptual metaphor theory and the embodied cognition account providing empirical evidence in support of them. Chapter 3 and 4 examine the body-specificity hypothesis, different studies in support of it and some theoretical explanations. Chapter 5 discusses the

stability of space-valence associations. Chapter 7 and 8 describe two experiments, which investigate the dependence of space-valence associations on the handedness and the reversibility of those associations. In Chapter 8, the results of the experiments and also the contribution to our knowledge to the representation of abstract concepts are discussed.

1. Conceptual knowledge representation

How do we store and organize the perceived information, and what are our thoughts made of? It is widely accepted that knowledge about a category (e.g., *dogs*) is organized in concepts. For decades, cognitive science tries to answer the question about the constitution of those concepts. There are two distinct directions in the scholarly literature in regards to the knowledge representation in the human brain - *amodal* and a *modal* one. Proponents of the amodal view suggest that the information perceived through different modalities is transformed and stored in a form of symbols in the semantic memory (e.g., [Marr, 1982](#); [Fodor, 1983](#)). Once the perceptual information is transformed into symbols representing, for example, feature lists, the sensory-motor system is no longer involved in those representations. Features operations are then performed on those new amodal representations (e.g., [Pylyshyn, 1999](#)). At the other end of the continuum are the proponents of the *grounded cognition* (e.g. [Barsalou, 1999, 2003, 2008](#); [Fischer & Coello, 2015](#)) who propose that our bodies determine the nature of our thoughts so that cognition cannot be considered separate from action and perception. “*Representations of external (perception) and internal states (proprioception, emotion and introspection) as well as actions critically constitute concepts and thus play a fundamental functional role in linguistic understanding and conceptual thinking*” ([Kiefer & Pulvermüller, 2011, p. 806](#)). In other words, perceived modal information is stored in form of mental representation which contains the information from the different modalities. Additionally, thinking about a concept involves activation of the

sensory-motor system. (Barsalou, 1999; Pulvermüller, 1999). For example, if one thinks about an apple, the visual system simulates the process of seeing the apple, the perceptual system simulates smell, taste, etc., and the motor system simulates grasping, biting, tearing, peeling etc. Thinking about an apple activates the same modality-specific brain areas which are activated when one is actually seeing an apple. There is no single unifying theory related to the grounding cognition, but there are different accounts providing an explanation to the process (see Barsalou, 2008 for a review). The current work will devote attention to two of the possible theories relevant for the propose of the study - the *conceptual metaphors theory* (Lakoff & Johnson, 1980, 2003) and the *theories of situated action* (Barsalou, 2003).

According to the *conceptual metaphors theory* (Lakoff & Johnson, 1980, 2003) people do not just use metaphors in the everyday speech they also think in metaphors. The authors suggest the existence of two types of concepts - non-metaphorical and metaphorical. Non-metaphorical concepts are those, with which we can have real experience: we can see them, touch them, etc. (e.g., *cat*). Metaphorical concepts, on the other hand, can only be understood in terms of other concepts. If we look at the metaphor “TIME IS MONEY”, the concept of “money” is something we can see, touch, and we can have experience with, whereas “time” can be understood only in terms of money or other concrete concepts (e.g., “TIME IS A FLOWING RIVER”). So the important thing in this theory is the mapping between the concrete source (*money*) and the abstract target (*time*) domains. The authors emphasized that metaphors are not of purely of linguistic origin but are a matter of thoughts, which is why they denote them “conceptual metaphors”. Lakoff & Johnson (1980, 2003) suggested three types of conceptual metaphors - orientational, ontological and structural ones. The orientational metaphor (“UP-DOWN”; “FRONT-BACK”) comes from the fact that our bodies are spatially situated. The idea is that concepts are spatially organized and related to each other – e.g., “GOOD IS UP; BAD IS DOWN” (*feeling up*). The ontological metaphors (e.g., “THE MIND

IS A BRITTLE OBJECT”) derive from our experience with objects and substances. “*They allow us to make sense of phenomena in the world in human terms—terms that we can understand on the basis of our own motivations, goals, actions, and characteristics*” (Lakoff & Johnson, 2003, p. 34). The third type, the structural metaphor allows for people to structure one kind of experience (*argument*) in terms of others (*war*) - (“ARGUMENT IS WAR”).

Proponents of the theory of situated action (e.g., Fischer & Coello, 2015; Wilson, 2002 for review), known also as *embodied cognition account* highlighted the importance of the interaction between our bodies, and the environment for the cognition. Our conceptual knowledge is grounded on the perceived features of the concepts and the interaction of our own body with those concepts. (Barsalou, 2008)

2. Empirical evidence of embodied concepts

There is evidence in support of the statement that concepts are grounded in concrete experience, that they are embodied. For instance, Pecher, Zeelenberg, & Barsalou, (2004) presented participants with words describing concepts of different objects paired with 3 names of properties, one of them being a target property. Properties could be of different modalities - visual (e.g., *brown*), motor action (e.g., *throw*), sound and touch. The authors reported faster reaction time (RT) for property verification when the concept was previously shown with property from the same modality rather than from a different one. As an explanation, they propose that when the concept occurs for the first time, a simulation (activation of the proper brain areas) relevant for the verification of the property runs, so when the concept occurs for the second time, the same modality is also activated. Further evidence comes from neuroimaging studies. For instance, Hauk, Johnsrude, & Pulvermüller, (2004) presented right handed participants with face-, arm- and leg-related action words. The functional magnetic

resonance imaging (fMRI) showed activation not only in regions responsible for visual word processing, but also in the left premotor cortex, which supported an embodied view with conceptual information distributed in different brain areas including those responsible for the motor action.

The studies described above are supporting an embodied view of the representation of concrete concepts (e.g., *dog*), concepts with which we can have bodily experience. The next chapter will discuss an embodied view of grounding abstract concepts

3. Challenges for the embodied cognition – abstract concepts

One challenge to the theory of embodied cognition are the abstract concepts such as *democracy*, *moral*, etc. How can we represent things we can never touch, smell, and see - concepts with which we do not have any bodily experience whatsoever? How can we ground them using sensory-motor simulations? A solution for an embodied representation of abstract concepts provides the *conceptual metaphor theory* [Lakoff & Johnson \(1980, 2003\)](#). As described above according to that theory “*our conceptual system is largely metaphorical, then the way we think, what we experience, and what we do every day is very much a matter of metaphor*” ([Lakoff & Johnson, 2003, p. 4](#)). The same goes for abstract concepts “*Abstract concepts are grounded metaphorically in embodied and situated knowledge*” ([Barsalou, 2008, p. 62](#)). We conceptualize abstract things in term of things we can experience through our sensory perception and motor action. For example, our bodily experience (e.g., *body posture when one is happy* – getting good grade) is grounded in conceptual mapping between concrete source domain (e.g., *space around the body*) and abstract target domain (e.g., *emotion*) concepts, in this case “GOOD IS UP”. According to the orientational metaphor the abstract concept of valence is organized spatially “GOOD IS UP”, “BAD IS DOWN”. Different

studies provide evidence that conceptual metaphors structure not only our representations of valence (good/bad) with the help of physical *space* (e.g., [Meier & Robinson, 2004](#)) but also *time* ([Casasanto & Boroditsky, 2008](#)), *number* ([Wood, Willmes, Nuerk, & Fischer, 2008](#) for review), etc. For example [Meier & Robinson \(2004\)](#) presented their participants with positive and negative words. The words appeared either on top of the screen or on the bottom. Participants were faster categorizing positive words shown on the top of screen, and negative words presented on the bottom than when presented with positive words on the bottom, and negative words on the top of the screen. The authors conclude that people automatically associate good things with ‘UP’ space around their bodies and bad things with “DOWN” space. That is, they automatically activate the conceptual metaphors “GOOD IS UP” and “BAD IS DOWN”. Further evidence for embodied representation of abstract concepts is provided by [Dreyer & Pulvermüller \(2018\)](#). The authors show that simply reading abstract emotional words activates in face- and hand-motor areas in the brain. The word activates not only semantic information, but also motor-information in line with the idea that our mental representation constitutes of sensory-motor experience.

To summarize, there is evidence that concepts representations emerge from the interplay between perception, motor action, proprioception, emotion, and the interaction of the own body with the environment. Taking into account the important role of our bodies, a question arises: do representations of those abstract concepts differ depending on our bodies? The next chapter introduces the body-specificity hypothesis ([Casasanto, 2009](#)) which advocates that mental representations depend on our bodies.

4. The body-specificity hypothesis (BSH)

Based on the theory of the embodied cognition which claims that thoughts consist of mental simulation of people's own bodily experience (e.g., Barsalou, 1999), Casasanto (2009) suggests that people with different bodies should also think differently. If thinking means mentally simulating and activating stored sensory-motor knowledge, then a tall person will simulate tearing an apple in a different way from a short person. The first one has probably to bend, and the second one has probably to lift. Imagine also that one of them is color-blind, so the mental representation of the color of the apple will be different related to the constraints of the own body. Besides, the one may be left- and the other right handed, and they perform the action using different hands, so that the neural activation will involve different brain areas. As a result, one and the same concept "*apple*" will have different mental representations depending on the people's bodies, therefore by thinking about one and the same concept, different mental simulations will arise and different brain areas will be active depending again on our bodies. Following that reasoning, Casasanto (2009) develops the body- specificity hypothesis (BSH), according to which people *„who interact with their physical environments in systematically different ways, should form correspondingly different mental representations”* (Casasanto, 2009, p. 351). In addition to Lakoff & Johnson (1980, 2003), who proposed that conceptual metaphors arise from linguistic, cultural and bodily experiences, Casasanto (2009) further investigated whether the bodily experience has a crucial role in the process of conceptualizing. He focused on a comparison between left- and right-handed people and their mental representations of abstract concepts, in particular, *valence*. Valence is an emotional value (positive vs. negative) associated with a current stimulus. Casasanto (2009) compared implicit mapping of abstract target domains like *good* and *bad*, and the concrete source domain, the lateral space around our bodies. In a fundamental

experiment ([Casasanto, 2009](#)), participants were asked to imagine that a person called Bob went to the ZOO (“Bob task”). There would be animals Bob liked and other he would not like. The participants were then supposed to draw (instead of Bob) the good and the bad animals in the left-right or up-down boxes ([Casasanto, 2009](#)). The results showed an expected effect of a vertical mapping, in line with the orientational conceptual metaphor GOOD IS UP and BAD IS DOWN ([Lakoff & Johnson, 2003](#); [Meier & Robinson, 2004](#)) with both right- and left-handed participants drawing the good animal in the top box and the bad animal in the bottom box. However, the results in the horizontal condition differed depending on the handedness. Most right-handed people draw the good animal in the right box, whereas most left handed draw it in the left one. In order to exclude the possibility that participants prefer drawing with the dominant hand in the dominant box, the author replicated the results using the same task but with only horizontal condition and asking the participants to respond orally with assigning the animals to the left/right boxes (Experiment 3, [Casasanto, 2009](#)). In two more experiments, participants were required to attribute positive and negative characteristics to people and fribbles (artificial three-dimensional stimuli figures with different shape, color, and texture). The results showed that the space-valence mapping differ depending on the handedness. Right-handed participants attributed more positive characteristics to fribbles/people presented on the right side whereas left- handed attributed more positive characteristics to images appearing on the left side (Experiments 4 and 5, [Casasanto, 2009](#)). The differences in the space-valence mapping between right-handers, and left-hander rule out the possibility that conceptual metaphors arise purely from linguistic or cultural experience, because both left-handed and right-handed people are exposed to the same linguistic and cultural influence. The results lead to the conclusion that mapping between valence and horizontal space depends on our bodies. According to [Casasanto \(2009\)](#), the more fluent interaction with the space around the dominant hand causes people to

implicitly associate their sensory-motor experience with positive emotions, and the disfluent interaction with the space around their subordinate hand with negative emotions. Thus, the body-specificity hypothesis connects motor fluency in a personal space with conceptualization of abstract concepts.

To conclude, [Casasanto \(2009\)](#) provides evidence that motor fluency shapes our abstract concepts which is in line with the embodied cognition account (e.g., [Barsalou, 2008](#)).

4.1 Evidence supporting the body-specificity hypothesis

Differences in the space-valence mapping associated with our bodies, were shown using different tasks, different stimuli (linguistic and nonlinguistic), different measurements methods (RT, oral judgment, free observation, fMRI), and across different languages and cultures (American, Dutch, Chinese, German, Moroccan etc.) (e.g., [de la Vega, de Filippis, Lachmair, Dudschig, & Kaup, 2012](#); [Kong, 2013](#); [de la Vega, Graebe, Härtner, Dudschig, & Kaup, 2015](#); [Casasanto & Jasmin 2010](#) ; [Casasanto & Henetz, 2012](#), [Casasanto 2011, 2014](#), for review). For example, different space-valence associations are found to be manifested in the early childhood. [Casasanto & Henetz \(2012\)](#) assessed the implicit and explicit space-valence associations by 5 and 10 years old children. In the first experiment participants should imagine a toy they like and another one they do not, and to place it in left or right box. In the second experiment they were presented with pairs of cartoon animals presented on both sides (left and right) of a page and were required to judge which animal was smart or dumb. Right-handed children put 54% of the good animals in the right box, and judged 59% of the animals presented on the right side as smarter. Left-handed children showed the opposite pattern assigning 83% of the good toy to the left side, and more positive characteristics to the 61% of the animals presented to their left side. Similar pattern observed [de la Fuente, Casasanto,](#)

Román, & Santiago, (2015). with Spanish and Moroccan adults. The researchers asked whether space-valence mapping could be influenced by the culture and especially by the writing direction. The result showed “*good is right*”, and “*left is bad*” mapping for both, Spanish as for Moroccan participants. In addition, Zhao, He, Zhang, Chen, Chen, & Huang (2016) found that first impression of people were influenced by the side they were presented. Participants were more likely to choose people who are better for date, or as friends, as boss or as national leader if they were presented on spatial location corresponding with ones dominant side.

Besides the Bob task and tasks required choices based on positive and negative (semantic) features, where valence was task-relevant, evidence for the body specific space-valence mapping was found outside the laboratory, and also using a memory task. It was established that during public speech, people used their dominant hand when gestures were accompanying a positive statement, and the other hand when the statement was negative. The effect was found for right-handers as well for left-handers. (Casasanto & Jasmin, 2010). In addition, Brunyé, Gardony, Mahoney, and Taylor (2012) found biases in spatial memory for positive information presented on the right side and negative on the left side for right-handers and the opposite pattern for left-handers. Participants were required to remember different locations on a map. Each location was accompanied by either negative or positive stories. When asked to recall the locations, right-handed people placed more positive events in the right part of the map, whereas negative events found place on the left side of the map. The left-handed did the opposite. Taken all together these studies support the body-specificity hypothesis in cases of task-relevant (Casasanto & Henetz, 012) and task-irrelevant space-valence relationship (Casasanto & Henetz, 2012; Brunyé et al., 2012).

While the research discussed above used off-line measurement, another line of research used on-line measurement (reaction time - RT) in an attempt to find “on-line”

support for the body-specificity hypothesis. [De la Vega et al., \(2012\)](#) measured RT asking participants to categorize positive and negative words. Participants were presented with emotional words, and non-words. The task was to respond only to the words by pressing left and right response buttons. Right-handed participants were faster responding to positive words by pressing a button with their right hand, and to negative words when responding with their non-dominant left hand. Left-handed participants showed the opposite pattern. The effect was observed only when the focus of the task was explicitly related to the emotionality of the words (judging if the word is positive or negative), but not in lexical decision task where valence was task irrelevant, and participants had just to differentiate between words and non-words. Interestingly, although there was no interaction when the task required lexical decision, main effect with faster responses for positive than for negative words was still there, which means that valence (although it was task irrelevant) was processed, but apparently not deep enough, probably, to cause space-valence mapping. In an analogues task, [Kong \(2013\)](#) observed similar effect with Chinese participants using linguistic stimuli (words), and non-linguistic ones (faces). People were faster responding to positive words and faces with their dominant hand, and for negative stimuli, with their subordinate hand. Similar results were observed by [De la Vega et al. \(2015\)](#) who asked participants to judge positive and negative words responding with their dominant leg.

The studies described above support the existence of an implicit space-valence association between positive emotion and the dominant side of our bodies and between negative emotions and the subordinate side of our bodies, but they do not rule out the possibility of the influence of the response position. [De la Vega, Dudschig, De Filippis, Lachmair, & Kaup, \(2013\)](#) tried to answer the question whether hand dominance or the spatial response coding is the crucial factor, making people to respond with crossed hands (right hand-left space, left hand –right space). The authors observed that *good is right/ bad is left* are

mapped on the response hand, rather than to the response button. Further, [de la Vega et al., \(2013\)](#) manipulated the instructions. In the first case participants were instructed to respond with their hands (focus on the hands) while in the second to press the given buttons (focus on the left/right space). This manipulation did not influence the results suggesting that explicit accentuation to the hand or the space did not influence the implicit space-valence associations. Similar results observed [Chen & Proctor \(2017\)](#) by comparing participants categorizing positive and negative words with crossed and uncrossed hands. Again the space-valence association was the same for both crossed hands (right hand was mapped to left response button and the left hand to the right one), and for uncrossed hands (reflecting the hand dominance rather than the response position). This suggests that valence is mapped depending on our bodies rather than merely on the space.

If concepts and word meanings are grounded in our own perceptions and actions, then their neural representations should also differ between right- and left-handed people. In an fMRI study ([Wilems, Hagoort & Casasanto, 2010](#)) both left- and right-handed participants completed lexical decision task on verbs related to actions performed with a hand (*to throw*) and non-manual actions (*to kneel*) followed by a task that required a mental imagery of those actions. The aim was to compare neural activation while reading the words and while explicitly imagining the action related to the given word. For manual action there was activation in the premotor cortex, more precisely, in Brodmann Area 6 which differed between right- and left-handers. Simple reading of action words activate the same area which are active during performing of this action. The right-handers showed stronger activation in the left hemisphere (LH), whereas in left-handers the activation was in the right hemisphere (RH). The same pattern of activation was observed during the mental imagery task but involving as well the premotor and motor cortex from the respective side. The researchers concluded that body-specific differences in the lexical decision task should be responsible for the contra-

lateral activation. Consistent with the BSH the authors conclude that “*people with different bodies, who perform actions in systematically different ways, use correspondingly different neural tissues for representing action verb meanings*” (Wilems et al., 2010, p. 72).

Thus overall supports the claim that conceptual knowledge representation depend on our bodies.

5. Stability of space-valence associations

All studies presented so far supported the body-specificity hypothesis. As explanation for the results, the authors suggest that the associations between space and valence originate from the experienced motor fluency during the interaction with the space around our dominant hand, and the experienced motor disfluency from the interaction with the space around the subordinate hand. Taking into account that most people have dominant hand and consequently experienced more fluency with the space around that hand, the concepts represented in such a way should be stable if not inborn. However, another line of experiments observed individuals with reversed handedness due to a stroke or a temporal handicap and shows that those associations can be easily reversed with reversing the motor fluency. Casasanto & Chrysikou (2011) explored the effect of “reversed handedness” on the space-valence mapping. Participants had either long-term changes in the motor fluency due to a stroke or induced short-term changes due to performing a fine motor task, arranging some dominoes wearing a ski glove (The “Domino task”). All participants then performed the “Bob task” (Experiment 3, Casasanto, 2009). For both groups the changed motor fluency corresponded with reversing the space-valence judgment. All participants were initially right-handed. Those whose left hemisphere was impaired, resulting in paralyzed right-hand, assigned the good animal to the left side. Also 63 % of those who wear the glove on their right hand (induced motor disfluency) assigned the good animal to the left box. The authors

concluded that “*motor experience is sufficient to determine the direction of space-valence associations*” (Casasanto & Chrysikou, 2011, p. 422). In addition, Kominsky & Casasanto (2013) and De la Fuente et al., (2015) showed that even experience of motor disfluency is not necessary. It is enough to observe somebody or imagine themselves performing a fluent or disfluent motor action. For example, Kominsky & Casasanto (2013) have examined whether taking another perspective would influence the space-valence associations. In previous experiments using the Bob task (Casasanto, 2009) participants had to judge how Bob would assign the animals to the boxes facing the same direction like Bob. In the current experiments participants were presented either with cartoon character’s Bob (abstract line drawing) or image of a real person “Bob” either from the same or the opposite perspective. They were asked to decide in which box the cartoon character’s /the real person would place the good and the bad animals. The results remained the same as in the previous studies. Participants have put the good animals on the cartoon character’s /man’s right side and the bad one on the left side when both faced in the same direction. In the opposite perspective condition, the majority of participants placed the *good* animal in the box on their own left. Interestingly, in the second experiment, a real person called “Bob” was wearing a sling on one of his hands. So the participants changed their judgment according to which hand was impaired. They assigned good animals according to the side to the healthy hand, and bad animals were placed on the side of the impaired hand. The experiments do not clarify the issue, whether participants computed the space-valence mapping taking the perspective from the other personality (allocentric) or from rotated egocentric view. An answer to that question was provided by de la Fuente, Casasanto, & Santiago, (2015). The authors divided the participants into actors and observers. Both were naïve to the aims of the experiment. In the first part of the experiment the actors performed the Domino task (Casasanto & Chrysikou, 2011) whereas the observer watched the execution carefully either face to face with the actor or from the same

perspective, sitting behind the actor. Secondly, during the test phase, they had to perform the Bob task adapted from Casasanto (2009). In the same perspective 83% of the actors who wore the glove on the right hand and 79% of the observers placed the good animal to the left like natural left-handers. The induced handedness phenomenon was observed both for actors and observers. In contrast, observers who had different spatial perspective from the actor showed opposite pattern of responses. When the actor wore the glove on the left hand, the observer placed the good animal to the left, whereas when the actor had the glove on their right hand, the most of the observers assigned the good animal to the right side. The authors concluded that *“the observers in Experiment 2 tended to spatialize “good” and “bad” on the basis of the fluent and disfluent actions they saw construed from their own egocentric spatial perspective”* (De la Fuente et al., 2015, p. 35). Further, Marsh, Kelly, Dark, & Oliver, (2013) aimed to show how differences in the visual, and the proprioceptive systems may contribute to space-valence association. Participants had an avatar hand which either moved like their own arm in the physical world (normal arm movement condition) or their avatar arm moved mirrored to their own arm (switched arm condition). They tried to induce motor disfluency using a task where participants were required was to touch some blocks with their avatar hand. Then the authors explored the space-valence associations using the Bob task (Experiment 3, Casasanto, 2009). When the avatar hand moved identical to the own hand (“normal” condition) normal “good is right” association was observed. The effect disappeared when participants were assigned to the “switched arm condition.” Although the authors interpret the results as importance of own proprioception, and not just the visual input, we can see that simple mismatch between visual input and arm movement cause space-valence associations to disappear. Importantly, later de la Fuente, Casasanto, Martínez-Cascales, & Santiago (2017) showed that kinesthetic experience is not necessary for reversing the effect. The simple imaging of themselves performing a motor action with an impaired hand caused participants

to reverse their space-valence associations. Similar results were observed for both, performing the Domino task or imagining performing that task. The authors concluded that “*our study appears to be the first to show that mental imagery, alone, can generate what Winkielman, Schwarz, Fazendeiro, and Reber (2003) called “the hedonic marking of fluency”*” (de la Fuente et al., 2017, p. 7).

Taking all together we can conclude that space-valence associations are not stable representations acquired during early child development. It depends on our momentary interaction (real or imagined) with the environment, or the observation of another person and the way she/he performs an action.

6. Rationale of the current work

It has been established that one implicitly associates the space around the dominant hand with positive valence, and the space around the subordinate hand with negative valence. The effect was observed using different stimuli, e.g. words, faces, and animals, also using different measurement technologies like RT (*categorizing emotional stimuli*) and fMRI (*representation of action verbs*), observed oral judgment (*the Bob task*), and simple free observation. The effect remains stable also across different languages and cultures (Casasanto 2011, 2014, for review). As explanation of that phenomenon an experienced motor fluency is suggested. The emotion accompanying the motor action makes one associate the more fluent interaction with the space around the dominant hand with positive experience, and the less fluent interaction with the space around the subordinate hand with negative one. Evidence for the importance of the experienced motor fluency (e.g., Casasanto & Chrysikou, 2011) comes from studies where different types of motor constraints were assessed resulting in

experiencing of disfluency of the actions carried out with the dominant hand. The changes in the space-valence associations were assessed only by using an off-line task (*The Bob task*). To the best of my knowledge, there is no previous research on the effect of the “reversed handedness” on the space-valence mapping measured with an on-line task, using RT technologies. The aim of the present research is to investigate whether changes in the motor fluency are strong enough to cause an on-line reversal in the implicit associations between valence and horizontal space using an RT categorization task. If so, this would indicate instability at least of the abstract concepts of valence. This will cast doubt whether abstract concepts generally can have stable mental representations or are the constructed „afresh” each time we use them.

7. Experiment 1 (RT valence word categorization)

7.1 Aims and Hypotheses

Experiment 1 aimed to replicate the results of [De la Vega et al., \(2012\)](#) and [Kong \(2012\)](#) that showed faster RTs for categorizing positive stimuli with the dominant hand and faster for categorizing negative stimuli with the subordinate hand both for left- and right-handed participants. As stimuli Bulgarian words classified as positive and negative were used. Generally, faster responses were expected when people had to respond with their dominant hand to positive words compared to negative ones and faster responses with their non-dominant hand to negative words than to positive ones. Previous research showed the existence of space-valence associations for depending on the handedness. However, to the best of my knowledge, there are no data that include ambidextrous. If the space-valence mapping is due to experienced motor fluency, then ambidextrous who should interact with

both hands equally fluent should not show any mapping between response hand and emotionality resulting in similar responses to positive and negative words for both hands.

Hypotheses

The main hypothesis is that space-valence associations depends on people's handedness. As a confirmation for this hypothesis, I expect the following results:

Right- and left-handed participants will respond faster to positive than to negative stimuli with their dominant hand and faster to negative than to positive stimuli with their subordinate hand, whereas ambidextrous will respond similar with both hands to positive and negative stimuli.

7.2 Method

7.2.1 Participants

In the experiment participated 117 (31 males) participants. Most of them were students at the NBU and participated voluntarily or in exchange for course credits. They all provided written informed consent for their participation. Their handedness was assessed with the NBU handedness inventory (for more details see below).

7.2.2 Stimuli and design

The stimuli material consisted of 96 abstract emotional Bulgarian words (48 positive and 48 negative). Seventy-two of them (e.g., “love/любов”, “grief/мъка”) were target words on which RT was measured. Sixteen (e.g., “harm/вреда”, “benefit/полза”) were used as “warm-up”-words, excluded from the data analysis, and 8 (e.g., “respect/уважение”, “rage/ярост”) as training-words (see Appendix A). The “warm-up”-words were presented at the beginning of each experimental block in order to give more time to the participants to get acquainted with the task and the response mapping. All target-words were matched for word length measured in number of letters, word frequency, and emotionality strength (see Table

1). Data of emotionality strength and valence were taken from unpublished data base in Bulgarian. Words were rated on a seven-point Likert scale with 1 - least emotionally strong/negative and 7 – most emotionally strong/positive. Two ratings were done separately with different participants: data of 24 participants per rating were gathered. Further, objective word frequency data was derived from a 72-million word data base (Simov et al., 2004), converted into frequency score per million and then 10-base logarithm of the score was taken with one added to the score per million to avoid the undefined $\text{Lg}(0)$.

Table 1. Means (M), standard deviation (SD), and standard error (SE) for the properties of the words used in Experiment 1 and 2. Words were rated using a 7-point Likert scale where 7 is the most positive and the emotionally strongest. Word length was measured in number of letters. All words were nouns, singular form.

	Positive words		Negative words	
	M(SD)	SE	M(SD)	SE
Frq (Log10)	1,30 (0,52)	0,09	1,14(0,43)	0,07
Word Length	6,00 (1,79)	0,30	6,14(1,96)	0,33
Emotionality	5,78 (0,36)	0,06	5,66 (0,34)	0,06
Valence	6,10 (0,35)	0,06	2,05 (0,38)	0,06

One-way ANOVAs did not show any differences between positive and negative words with regard to their frequency [$F(1, 71) = 1.96, p = .166$], length [$F(1, 71) = .99, p = .755$], and emotionality [$F(1, 71) = 2,11, p = .151$], and did show significant difference with regard to their valence [$F(1, 71) = 2237,72, p = .000$].

Experiment had 3 x 2 x 2 mixed factorial design with two within-subject factors, valence (positive vs. negative) and response hand (left vs. right) and one between-subject factor (handedness: right vs. left vs. ambidextrous). The dependent measure was the reaction time for categorizing the words as either positive or negative.

7.2.3 Procedure

Participants took part in the experiment individually. They were seated in a small booth facing the 17-inch LCD monitor with 1024×768 resolution from 50-52 cm distance. Written instructions (see Appendix B) appeared on the monitor and participants were informed that an emotional word will appear on the screen and they should answer as fast as possible for positive and negative words with either “z” (left hand) or “3” (right hand). At the start of each trial a black fixation cross was presented over a white background for 500 ms following by either positive or negative word presented in the middle of the screen. All stimuli appeared in black, 24 pt., Courier fonts, against white background. The word remained displayed on the screen for 3000 ms or until a subject’s response was given. Intertrial interval was set to 1000 ms. The experiment was divided in two blocks in order to counterbalance the response positions. Response buttons were counterbalanced within-subject so that each subject pressed half of the trials with their right hand for positive, and with their left – for negative, and the other half – with the opposite hands. Additionally, the order of button mapping was also counterbalanced so that half of the subjects started with right hand mapped with positive, and the left hand – with negative words. The other half of the subjects started with left response button for positive words and right response button for negative one. Before the beginning of each block, participants had a practice session with 8 words that did not appear in the experimental part. The 8 words were repeated twice (16 practice trials). The experimental part contained 36 target-words and 8 “warm-up”-words presented at the beginning of each experimental part. Practice trials and “warm-up”-trials were later excluded from the RT analysis. The experiment was created with E-prime 2.0 ([Schneider, Eschman, & Zuccolotto, 2002](#)). After finishing the word categorization task the participants filled in the NBU handedness inventory (see Appendix C). It consists of 20 items and 3 open questions. The items describe tasks which one has to imagine performing. Participants were required to mark

the most probable hand, foot or eye with which they are going perform the tasks described in the items. The individual score was calculated based the averaged item-scores. The individual result could take values in the range between 1 and 3 points. As right-handed were classified people with a score between 2,5 and 3 points, participants with score between 1.75 and 2.25 were categorized as ambidextrous, and with score below 1.5 as left-handers. Between each category, right-handed, left-handed and ambidextrous 0,5 points (score between 1.6-1.7 and 2.3-2.4) were left as undefined for participants who could not clearly be assigned to one of the pre-defined groups.

The experiment took about 12 minutes to execute.

7.3 Results and discussion

A total of 26 subjects (22 %) were excluded from the data. Data of fifteen of them tell outside of the predefined ranges for the three handedness groups. Three more subjects were excluded because they did not complete the whole inventory, one because of problems with the eyes and two because of accuracy under the estimated threshold of 90%. Five subjects have been participating twice, so their second attempt was excluded. From the remaining 91 participants (23 males), 55 were classified as right-handed, 19 - as left-handed, and 17 - as ambidextrous. Their age ranged from 18 to 43 years, with a mean age of 25 years ($SD = 5$).

The following data trimming procedure was used: trials in which participants made an incorrect response (2.9 % of the data) were excluded. Also, trials in which RTs fell outside of mean $RT \pm 2SD$ per conditions were excluded (4.6% of the remaining data). Error analysis was not performed, because of the overall high level of accuracy, above 97%. RTs were averaged by subject and data were submitted to a repeated measures ANOVA (see Table 2) with two within-subject factors (valence: positive vs. negative and response hand: left/right) and one between-subject factor (handedness: left vs. right vs. ambidextrous).

Table 2. Descriptive statistics per condition. Mean RTs (ms), standard deviations (SD) and 95% confidence intervals (CI), averaged by subject.

	NEGATIVE WORDS				POSITIVE WORDS			
	Left Hand		Right Hand		Left Hand		Right Hand	
	M(SD)	CI(95%)	M(SD)	CI(95%)	M(SD)	CI(95%)	M(SD)	CI(95%)
Right-handers	715 (14)	668-742	779 (15)	749-809	738 (13)	712-765	668 (13)	643-694
Left-handers	780 (23)	734-825	750 (26)	699-800	691 (23)	646-736	715 (22)	672-756
Ambidextrous	687 (24)	639-736	692 (27)	638-745	659 (24)	611-707	620 (23)	575-666

The three-way interaction between valence, response hand and handedness was statistically significant [$F(2, 88) = 15.351, p = .000, \eta_p^2 = .25$] (see Figure 1). The Bonferroni post-hoc test revealed that right-handers responded faster to positive stimuli with their right hand than with their left hand (668 vs. 738 ms) and faster to negative stimuli with their left hand compared to the right one (715 vs. 779 ms). The only significant post hoc result observed by left-handers was the faster RT for positive than for negative stimuli when responding with a left hand (691 vs. 780). Ambidextrous did not show any significant post-hoc result. Further there was statistically significant two-way interaction between valence and response hand [$F(2, 88) = 7.053, p = .009, \eta_p^2 = .07$]. The two-way interactions between handedness and valence [$F(2, 88) = 1.135, p = .326, \eta_p^2 = .02$], and between handedness and response [$F(2, 88) = 0.921, p = .401, \eta_p^2 = .02$] position were not significant. A main effect of valence was found [$F(2, 88) = 90.231, p = .000, \eta_p^2 = .56$] with faster RT for positive (665 ms) than for negative (733 ms) words and a main effect of handedness was found [$F(2, 88) = 3.311, p = .041, \eta_p^2 = .04$] with faster RT observed by ambidextrous participants (665 ms), compared to right-handers (725 ms) and left-handers (710 ms). The main effect of the response hand was not significant [$F(2, 88) = 2.743, p = .101, \eta_p^2 = .03$]

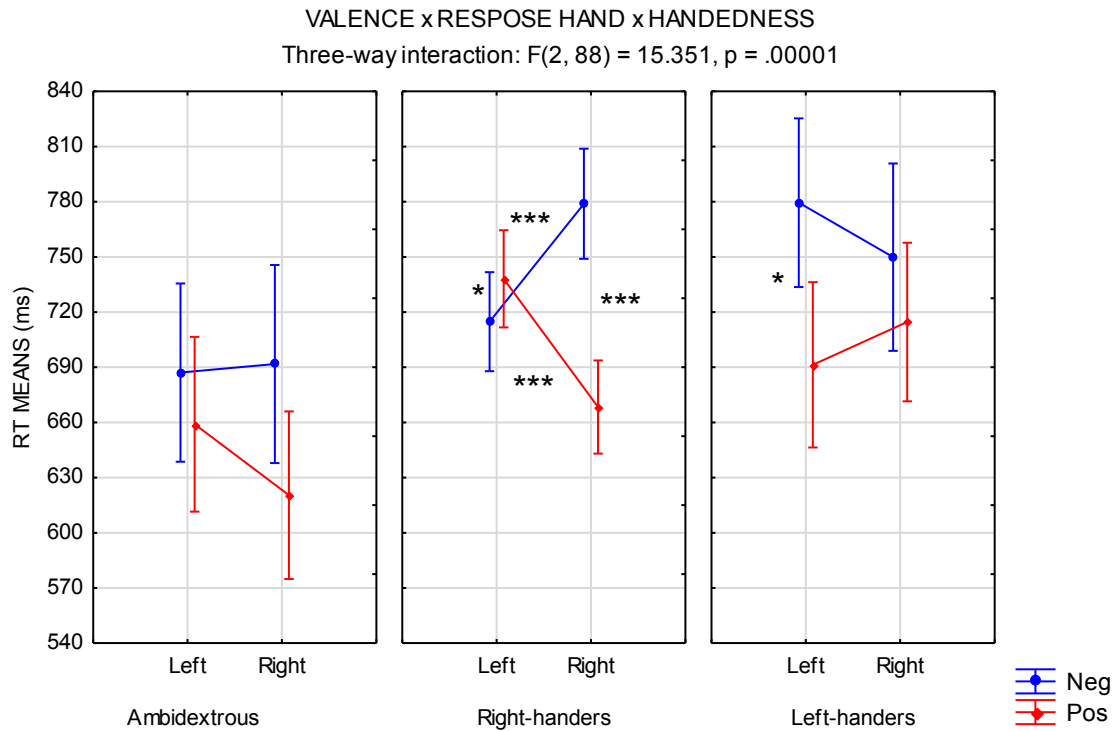


Figure 1. Interaction between valence, response hand and hand-dominance * - $p < 0.05$, *** - $p < 0.001$, Bonferroni post-hoc.

The aim of the study was to investigate whether the associations between space and valence differ depending on the hand dominance. Three groups of participants (left-handed, right-handed and ambidextrous) performed a valence word categorization task. Positive and negative responses were mapped to the left/right hand, and vice versa. In line with previous research (e.g., De la Vega et al., 2012) the results showed statistically significant three-way interaction revealing that the space-valence associations differ depending on the handedness of the participant. Right-handed subjects showed “positive is right” and “negative is left” mapping. The RT profiles by the left-handed participants showed the opposite RT tendency (“positive-left” and “negative-right”). Although the interaction between space and response hand observed by left-handers was not significant, in line with previous research (De la Vega et al. 2012; Kong 2012) left-handers responded significantly faster to positive than to negative words with their left hand. The effect for the right hand was weaker with no significant

difference between positive and negative responses. A reason for that can be that left-handed people are living in a world shaped mostly for right-handed. So they are forced to use their non-dominant right hand more often, which maybe cause fluency to spread also to their subordinate hand. Ambidextrous did not show any significant results. Thus, for them there was no strong space-valence association. The results suggest that the more fluent interaction of both hands causes the space valence mapping to disappear.

It needs to be said that previous research found a significant space-valence interaction in the left-handers group only when analyzing by-item, but not by-subject ([De la Vega et al. 2012](#); [Kong 2012](#)). Unfortunately, a by-item analysis could not be carried out because of a technical error in the script, so I cannot have direct comparison between my results and previous observed results by left-handed participants. Moreover, the interpretation of by-item analysis of linguistic stimuli can be misleading ([Clark, 1973](#)).

In order to achieve better understanding of the effects, and to be able to generalize beyond the concrete stimuli and subjects, a Linear Mixed Model (see Appendix D) with valence, response hand, and handedness as fixed factors and subject and items as random factors were conducted. The results revealed that the model including a three way interaction term accounts better for the data.

To conclude, the experiment supported the results provided by [De la Vega et al., \(2012\)](#) and [Kong \(2012\)](#) showing that space-valence associations differ depending on handedness. “Good is right” and “left is bad” associations were observed in right-handers, opposite tendencies were observed in left-handers. Further the results was extended to ambidextrous participants who did not show associations between space and valence.

8. Experiment 2 (Motor fluency and space-valence associations).

8.1. Aim and Hypotheses

The aim of the second experiment was to further investigate to what extent the currently experienced motor (dis)fluency is a crucial factor for the space-valence mapping. Previous research showed that space-valence mapping could be reversed by an experience of motor disfluency or by just observing somebody performing a disfluent action. To the best of my knowledge, the entire current research on the stability of the space-valence mapping relies on an offline experimental procedure (The Bob task). Although significant, the observed effects are weak, e.g. around 60 % of the right-handers reversed their mapping to “good is left”. In my opinion there are many problems with that approach. First, experiments that showed motor change due to stroke and/or some sort of different short-term handicap were carried-out on a very small sample size. Besides, the Bob task consists of only one trial which make it difficult, even impossible, to make more than speculative conclusions about the stability of the space-valence mapping. In order to make a general conclusion about an effect, a more precise processing-based measure is needed. The aim of the current experiment was to test the alternation of the space-valence association due to a shortly induced motor disfluency. Exactly the same Domino task adopted from [Casasanto & Chrysikou, \(2011\)](#) was used. But in addition to a Bob-similar task (assigning good/bad person to left/right box) RT for categorizing positive and negative words (same task and stimuli used in Experiment 1) was also measured. If the body-specific space-valence association is influenced by the experienced motor disfluency, then right-handed people who wore the glove on their left hand should show the same pattern of results observed by the right-handed participants in Experiment 1 (faster responses to positive words with the right than with the left hand, and faster responses to

negative words with the left than with the right hand). Right-handers who had the glove on their dominant hand should show opposite space-valence mapping similar to the left-handers observed in Experiment 1 (faster responses to positive words with the left than with the right hand, and faster responses to negative words with the right than with the left hand).

Hypotheses

The main hypothesis is that currently experienced motor disfluency will reverse the space-valence associations. As a confirmation for this hypothesis, I expect the following results:

1) Right-handers who wore the glove on their right hand during the Domino task were supposed to experience reversal of their initial space-valence associations in “good is left” and “bad is right ” mapping

- They will assign a person they like to the left box and those they do not like to the right one;
- They will be faster in responding to positive stimuli than to negative ones with their left hand and faster to negative stimuli than to positive ones with their right hand (the opposite pattern observed for right-handers in Experiment 1).

2) Right-handers who wore the glove on their left hand during the Domino task were supposed to keep their initial space-valence associations - “*good is right*” and “*bad is left*” unchanged.

- They will assign a person they like to the right box and those they do not like to the left one;
- They will be faster in responding to positive stimuli than to negative with their right hand and faster to negative stimuli than to positive ones with their left hand (the same pattern observed in experiment 1 is expected). They also will assign the good person to the right box and the bad person to the left one.

8.2 Method

8.2.1 Participants

Overall there were 75 participants. All of them were NBU students and voluntarily or in exchange for course credits. They all provided written informed consent for their participation. Their handedness was assessed with the NBU handedness inventory (Appendix C)

8.2.2. Stimuli and designs

The experiment consists of three parts - the Domino Task, the Person Task (described below in details) and the exact same emotional-word-categorization task used in the first experiment.

The Person task had 2 x 2 x 2 factorial design with two within-subject factors (person: like vs. dislike) x 2 (box: left vs. right) and one between-subject factor (glove: right hand vs. left hand). The dependent measure was observed frequencies for each of the eight conditions.

The emotional word categorization task had 2 x 2 x 2 mixed factorial design with two within-subject factors - valence (positive vs. negative) and response hand (left vs. right) and one between-subject factor glove (right hand vs. left hand).

8.2.3 Procedure

All of the participants performed firstly a fine-motor task (*The Domino task*) adopted from Casasanto & Chrysikou (2011) followed by an oral judgment task (*The Person task*), which was similar to the task used by Casasanto & Henetz (2012) and finally, the same valence word categorization task as in the first experiment.

1) First part - Domino task

Participants were seated on a 120 x 55.5 cm table. Forty green and forty orange rectangle stickers were placed on the table, forming 8 rows, each of which had stickers, placed approximately 12 cm apart from each other. The domino pieces were placed in a single box in the center of the table (see Figure 2). Participants were given the written instruction to place the dominoes on the squares using both hands simultaneously. In order to induce motor disfluency participants had to wear a bulky ski glove either on their right or on their left hand. The second glove was attached to the first in order to make the task more difficult. When the subject moved his/her hand, the second glove hanged down. The task was to arrange the dominoes upright placing them on the shorter side with the points facing to the participant. All people were required to start with the top row and the outermost rectangle, completing the task row by row. If a domino fell down, participants had to bring it back up before moving to the next domino. They were not allowed to assist the gloved hand with the ungloved one or vice versa. Participants had 12 minutes to complete the task. In case they were ready earlier, they had to put the dominoes back in the box again using both hands and to start arranging them again if any more time was left.



Figure 2. The Domino Task

2) Second part - The Person Task

Immediately after finishing the 12 minute Domino task, the participants were asked to move to an experimental booth. On a table in front of them, two empty boxes were placed, apparently 60 cm apart, to the left, and to the right hand side of the participants. Participants received written instructions (see Appendix B), on which two self-adhesive memo pads (a blue one and a green one) were glued. The participants had to imagine someone they liked and someone else they did not like. Then they had to write the first letter of each name on each sheet and then put both sheets in one of the two boxes placed on the right and on the left sight. Half of the participants wrote the good person on the blue sheet and the bad on the green one. The other half did the opposite. Also, the instruction for half of the subjects started with the good person first, the other half of the participants were required to start with the bad one.

3) Part 3 -word categorization task

After finishing the Person task participants performed exactly the same task used in Experiment 1. There the participant had to categorize words as positive or as negative by responding as fast as possible with either “z” (left hand) or “3” (right hand).

The whole experiment took about 25 minutes.

8.3 Results and discussion

8.3.1 The Person-task

From the 75 participants, 4 (5%) were excluded. One subject was excluded due to assigning both persons to the right box, and three more subjects - due to being ambidextrous. The remaining 71 subjects were all assessed as right-handed with a score between 2.45 and 3. The age range was between 17 and 50 year, with an average age of 23 years (SD = 6).

Table 3. Number of sheets with the liked / disliked person placed in the left vs. right box from people wearing

the glove on the right hand or the left hand during the Domino task.

GLOVE	RIGHT HAND		LEFT HAND	
	Like	Dislike	Like	Dislike
Left box	16 (46%)	19 (54%)	21 (58%)	15 (41%)
Right box	19 (54%)	16 (46%)	15 (41%)	21 (58%)

A chi-square test for associations between person (like vs. dislike) and box (right vs. left) was conducted (see Table 3). There was no statistically significant association between person and box position when participants previously performed the Domino task with glove on the left hand [Pearson's $\chi^2(1, N = 35) = 2.649$, $p = .104$; Fisher's exact test $p = .163$, two-tailed], and no statistically significant association when they performed the Domino task with glove on the right hand [$\chi^2(1, N = 36) = 0.486$, $p = .495$; Fisher's exact test $p = .642$, two-tailed].

The aim of the Person task was to explore the reversibility of the space-valence association using an off-line task similar to that used by [Casasanto & Chrysikou \(2011\)](#). The chi-square test revealed that there were no associations between the space around and valence neither in the group where participants wore the glove on the right hand (reversed handedness), nor in the group where participants wore the glove on the left hand (perceived handedness). The achieved results contraindicated the results observed by [Casasanto & Chrysikou \(2011\)](#) where participants wearing the glove on the right hand showed *positive is left, negative is right* mapping. Whereas participants whose left hand was gloved showed the opposite, *good is right, bad is left* mapping typically observed by left-handers. To conclude, the Domino task could not reverse the „off-line” space-valence associations, but at least caused them to disappear, probably.

8.3.2 RT categorization task

From all of the 75 subjects, three were excluded because of their being ambidexterous, one was not a native speaker, one had confused the response mapping, and three were excluded because of lower accuracy-under 90%., and two because of many RT outliers (more than 40% of all trials). The remaining 65 (87%) participants (14 males) were all categorized as right-handed with a score between 2.5 and 3. They ranged in age from 17 to 50 year, with an average age of 23 years (SD = 6).

The following data-trimming procedure was used: trials in which the participant made an incorrect responses (3.4 % of the data) were excluded. Also, trials in which RTs fell outside of mean RT $\pm$ 2SD per conditions were also excluded (4.9% of the remaining data). Error analysis was not performed, because of the overall high level of accuracy, above 96%. The RTs were averaged by subject and were then submitted to a repeated measures ANOVA (see Table 4) with two within-subject factors (valence: positive vs. negative and response hand: left vs. right) and one between-subject factor (glove: left vs. right).

Table 4: Mean RTs (ms), standard deviations (SD) and 95% confidence intervals (CI) for each condition averaged by subject. Experiment 2.

NEGATIVE WORDS					POSITIVE WORDS			
	Left Hand		Right Hand		Left Hand		Right Hand	
	M(SD)	CI(95%)	M(SD)	CI(95%)	M(SD)	CI(95%)	M(SD)	CI(95%)
Glove_LEFT	751 (20)	712-791	761 (19)	723-800	734(18)	698-770	706 (19)	668-744
Glove_RIGHT	808 (19)	767-847	779 (19)	741-817	746 (18)	710-782	743 (19)	706-781

The three-way interaction was not significant [$F(1, 63) = 2.852$, $p = .096$, $\eta_p^2 = .04$]. All of the explored two-way interactions: between valence and response hand [$F(1, 63) = 0.089$, $p = .768$, $\eta_p^2 = .00$], between valence and glove hand [$F(1, 63) = 0.946$, $p = .335$, $\eta_p^2 = .01$], between response position and glove hand [$F(1, 63) = .320$, $p = .574$, $\eta_p^2 = .00$] did not reach

statistical significance. A main effect of valence was found [$F(1, 63) = 48.744, p = .000, \eta_p^2 = .44$] with faster RT for positive (732 ms) than for negative (775 ms) words. The main effect of response hand was also significant [$F(1, 63) = 4.486, p = .038, \eta_p^2 = .006$] with faster RT for responses with the right hand (747 ms) than with the left one (760 ms). The main effect of glove position [$F(1, 63) = 1.707, p = .196, \eta_p^2 = .02$] was not significant.

In order to be able to generalize beyond the concrete stimuli and subjects, a Mixed Linear Model with valence (positive vs. negative), response hand (left vs. right), and glove (left hand vs. right hand) as fixed factors and subject and items as random factors were conducted. The linear mixed model revealed similar pattern of results (see Appendix D).

The RT categorization task aimed at investigating the influence of the motor (dis)fluency on the space-valence associations of right handers. Previous research ([De la Vega et al., 2012](#); [Kong 2012](#)) and also experiment 1 in the current work demonstrated the “positive is right” and “negative is left” space valence associations by right-handed participants. Further, it was established that those associations could be easily reversed (off-line) by inducing a short-term motor disfluency ([Casasanto & Chrysikou, 2011](#)). The currently observed data could not confirm the reversibility of the associations observed previous using an off-line task. However, the current on-line data showed that the induced motor disfluency leads the association to disappear. Right-handers who wore the glove on their right hand (reversed handedness) or on their left hand (perceived handedness) did not show space-valence associations anymore.

In both experiments (word categorization before/ after inducing of motor disfluency) a main effect of valence was observed suggesting that the effect of faster identification of positive words was not influenced by motor disfluency. Further participants wearing the glove on the right hand showed faster RT for positive than for negative stimuli only when responding with their left hand (746 vs. 808). These results profiles look similar to the results

of the left-handed participant observed in the first experiment.

To conclude, the fine motor task (The Domino-Task) caused the space-valence associations in right-handed participant to disappear when they wore the glove either on their left or their right hand. For the first time it was showed that the processing of the abstract concepts changed when motor fluency become different. In the whole previous research, conclusions about the process were not possible, because those associations were assessed only with the Bob task, which is related to decision making (off-line measure) rather than information processing (on-line measure).

8.4 Additional analyses

1) Comparison between right-handed participants in Experiment 1 and participants wearing the glove on their right hand in Experiment 2

In order to get more precise information about the influence of the induced motor disfluency the data of the right-handed participants in experiment 1 and the data of the participants who wore the glove on their right hand in experiment 2 were combined, and analyzed. The data were submitted to a 2 x 2 x 2 repeated measure ANOVA with valence (positive vs. negative) and response hand (left vs. right) as within-subject factors and glove (no glove vs. glove on the right hand) as a between subject factor.

Table 5. Mean RTs (ms), standard deviations (SD) and 95% confidence intervals (CI) for each condition averaged by subject

Right-handers	NEGATIVE WORDS				POSITIVE WORDS			
	Left Hand		Right Hand		Left Hand		Right Hand	
	M(SD)	CI(95%)	M(SD)	CI(95%)	M(SD)	CI(95%)	M(SD)	CI(95%)
No GLOVE	715 (15)	686-744	779(16)	748-810	738 (14)	748-810	668 (13)	642-697
GLOVE_RH	808 (19)	767-847	779 (19)	741-817	746 (18)	710-782	743 (19)	706-781

The three-way interaction was significant [$F(1, 86) = 30.104, p = .000, \eta_p^2 = .26$] as shown in Figure 3. The Bonferroni post-hoc test revealed that right-handers who did not perform the Domino-task responded faster to positive stimuli with their right hand than with their left (668 vs. 738 ms), and faster to negative stimuli with their left hand compared to the right one (715 vs. 779 ms). The only significant differences were established for participants wearing the glove on the right hand with faster RT for positive than for negative stimuli only when responding with a left hand (746 vs. 808). Statistical significance of a two-way interaction was accepted at a Bonferroni-adjusted alpha level of 0.05. From the two-way interactions, the interaction between response hand and valence [$F(1, 86) = 13.832, p = .000, \eta_p^2 = .14$] was also significant. The other two, those between valence and experiment [$F(1, 86) = 0.203, p = .653, \eta_p^2 = .003$] and between response position and experiment [$F(1, 86) = 1.526, p = .220, \eta_p^2 = .017$] were not significant. A main effect of valence was found [$F(1, 86) = 66.477, p = .000, \eta_p^2 = .44$] with faster RT for positive than for negative words and a main effect of experiment [$F(1,86) = 4.379, p = .039, \eta_p^2 = .04$], with faster RT observed by right-handed participants in experiment 1 compared to participant who wore the glove on their right hand experiment 2. The main effect of response position was not significant [$F(1, 86) = 3.160, p = .079, \eta_p^2 = .003$].

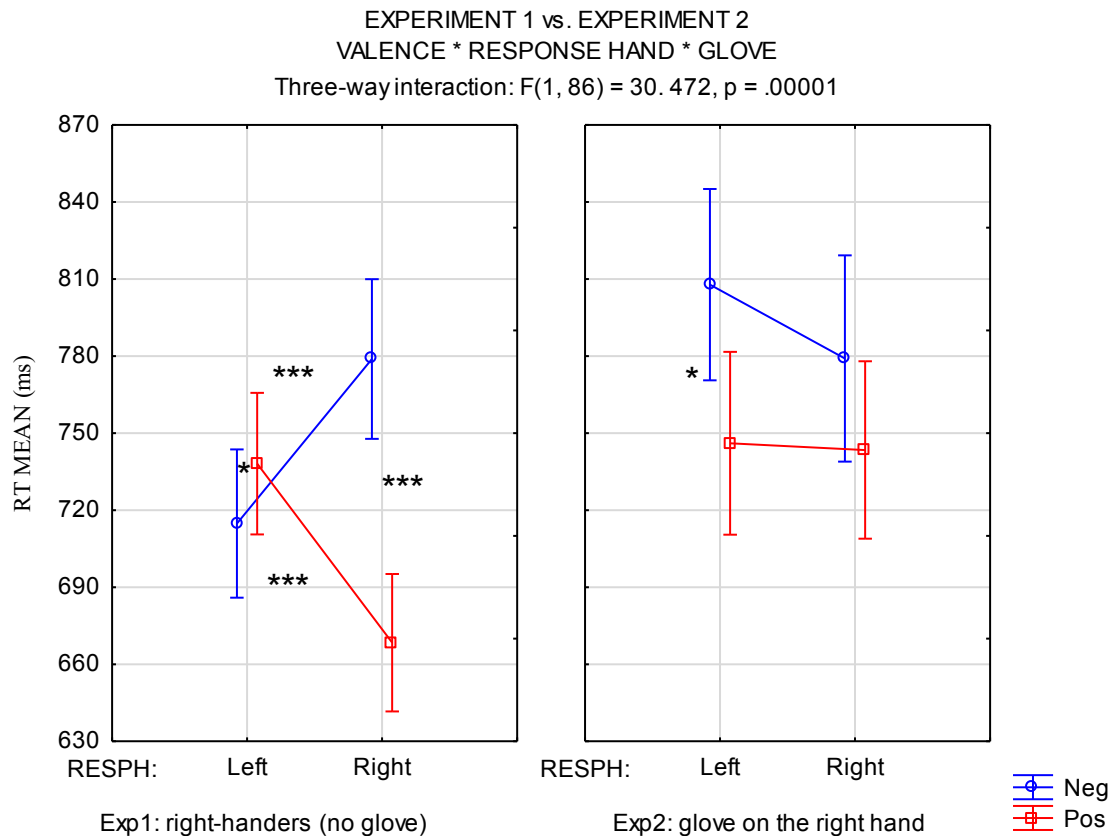


Figure 3 Interaction between valence, response hand, and subject group - right-handed from Exp. 1, and right-handed after the Domino task with the glove on their right hand (Exp. 2). * - $p < 0.05$, *** - $p < .001$, Bonferroni post-hoc

The significant three-way interaction showed that the fine motor task influenced the associations between space and valence. The disfluent motor interaction with the right hand lead to changes in the space-valence mapping. The effect for positive stimuli disappeared completely, participants were equally fast responding to positive stimuli with both hands (746 vs. 743 ms). They responded to negative stimuli faster with the right hand than with the left hand (779 vs. 807 ms).

2) Comparison between left-handed participants in Exp1 and participants wearing the glove on their right hand in Exp 2

According to Casasanto & Chrysikou (2011) people that wore the glove on their right hand should reverse their association to a pattern similar to natural left-handers. In order to

further explore this, the results observed by the left-handers in Experiment 1 were compared with the results of the right-handed participants who wore the glove on their right hand (reversed handedness condition) in the second experiment (see Table 6).

Table 6. Mean RTs (ms), standard deviations (SD) and 95% confidence intervals (CI) for each condition averaged by subject

	NEGATIVE WORDS				POSITIVE WORDS			
	Left Hand		Right Hand		Left Hand		Right Hand	
	M(SD)	CI(95%)	M(SD)	CI(95%)	M(SD)	CI(95%)	M(SD)	CI(95%)
No GLOVE (left-handers)	780 (23)	734-825	750 (26)	699-800	691 (23)	646-736	715 (22)	672-756
GLOVE_RH	808 (19)	767-847	779 (19)	741-817	746 (18)	710-782	743 (19)	706-781

The data was submitted to a 2 x 2 x2 repeated measure ANOVA with valence (positive vs. negative) and response hand (left vs. right) as within-subject factors, and glove (no glove vs. glove) as a between-subject factor.

The three-way interaction was not significant [$F(1, 50) = 0.413$ $p = .523$, $\eta_p^2 = .00$]. as shown in Figure 6. All two-way interactions, the valence-glove [$F(1, 50) = 0.654$ $p = .419$, $\eta_p^2 = .01$] and response hand- glove [$F(1, 50) = 0.628$ $p = .432$, $\eta_p^2 = .01$] ones in particular but also the interaction between valence and response position [$F(1, 50) = 3.599$ $p = .063$, $\eta_p^2 = .06$] were not significant. Again a significant main effect of valence was obtained. [$F(1, 50) = 4.794$ $p = .000$, $\eta_p^2 = .49$]. The main effects of response position [$F(1, 50) = 1.422$ $p = .238$, $\eta_p^2 = .02$] and of glove [$F(1, 50) = 1.632$ $p = .207$, $\eta_p^2 = .03$] were not significant.

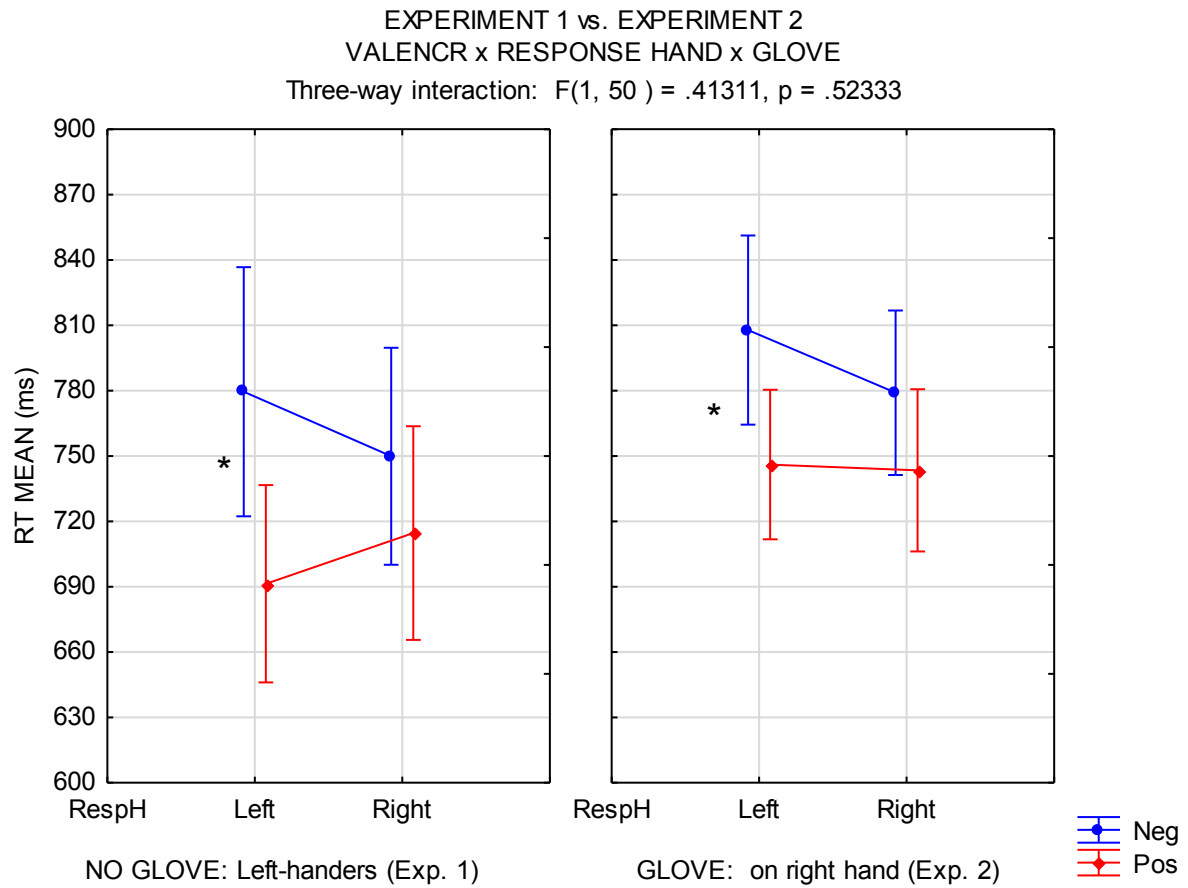


Figure 4. Interaction between valence, response hand, glove - left-handed from Exp 1 without glove and right-handed after the domino task with the glove on their right hand (Exp. 2).

In conclusion, the induced motor disfluency causes the right-handers' RT profiles to change and to look like those observed by "natural" left-handers in experiment 1 (see Figure 4).

9. General Discussion and Conclusion

The aim of the current research was to investigate how abstract concepts are represented in the mind, and to what extent are those representations influenced by the interaction of our own body with the environment. The investigation was inspired by the body-specificity hypothesis proposed by Casasanto (2009). According to that hypothesis, people have different mental representations of abstract concepts as a result of the peculiarities of their bodies. A typical example is handedness. Between 75% and 90% of the people are right-handed (Corbetta & Williams, 2006). There is still debate about the origin of the handedness. However, regardless whether handedness is due to sensory-motor experience or it has genetic roots, the consequence is that people tend to perform most of the tasks with their dominant hand which results in more fluent interaction not only with that hand, but also with the whole space around that hand. The experienced motor fluency associated with an action is positively (hedonically) marked (Winkelmann & Cacioppo 2001; Winkielman, Schwarz, Fazendeiro, & Reber, 2003) which leads people to associate the fluency in the motor action of the dominant hand with positive affect and the disfluency in the motor action of the subordinate hand with negative affect. For example, Hayes, Paul, Beuger and Tipper (2008) asked people to move neutral objects - either fluently or non-fluently due to an obstacle. Afterwards, people preferred more the objects that have been moved fluently, emphasizing the positive affect as a consequence of the experienced motor fluency. To summarize, the experienced motor fluency with the space around the dominant hand caused left-handers to associate positive affect with the left space, and right-handers - to associate positive affect with the right space. These associations are manifested in faster RTs when responding to positive words with the dominant hand and with faster RTs for negative words when responding with the subordinate hand (De la Vega et al., 2012; Kong, 2012). In line

with that research experiment 1 aimed at replicating the previous outcomes for categorizing positive and negative words by left-handers, right-handers, and to extend the results including ambidextrous participants. Overall, participants were faster in categorizing positive stimuli compared with negative ones. The results are consistent with the automatic vigilance hypothesis, according to which the more attention is paid to negative stimuli the slower the responses to negative stimuli compared to responses to positive or neutral ones are (e.g. [Algom, Chajut, & Lev, 2004](#)). Similar to the results observed by [De la Vega et al., \(2012\)](#) and [Kong, \(2012\)](#), right-handers showed significant space-valence association, responding faster to positive stimuli compared to negative stimuli with their right hand and also, responding faster to negative stimuli compared to positive stimuli with their left hand. Left-handers showed the opposite RT tendency. The mean RT differences by left-handers were much smaller and I suggest that this fact can be related to different reasons. First, the left-handers (N=19) sample was too small compared to right handers (N=55) which can influence the effect magnitude. Second, we know that left-handers are forced to use more often their non-dominant right hand, because most of the people are right-handers so most of the devices in the environment are adapted for right hands. Probably, this can lead to increasing the experience of fluency also with the non-dominant right hand by left-handers. So if we assume that space-valence associations are a consequence of experienced motor fluency as suggested by Casasanto (2009), and that left-handers experience fluency also with the non-dominant hand, then effect of the space-valence association will probably get weaker. This is also in line with experiments showing that left-handers used their dominant hand less than right-handers and also show less functional asymmetries than right-handers (e.g. Kasai, & Yahagi, 1999).

Another interesting case is the ambidextrous people who are supposed to interact equally fluent with both hands. As far as I know in the current research, such experimental group is explored for the first time in this research field. Ambidextrous participants showed overall

faster RTs responding to positive words than to negative ones but no interaction between valence and response hand was found. The results suggest that the fluent interaction with both hands cause the space-valence effects to disappear. But still the RT-profiles were similar to right-handers. This could mean that either ambidextrous are performing more tasks with their right-hand or that there is another explanation for the space-valence association in addition to or completely different from the motor fluency.

Another speculative explanation beyond the motor fluency is an attentional account which suggests that people pay different amount of attention to the space around their dominant hand compared to the space around their subordinate hand ([Peters, 1981](#)). The more often use of the dominant hand probably cause people to devote more attention to that side. It has to be said that there is no unambiguous opinion whether the attentional bias is a consequence or cause of handedness ([Buckingham & Carey, 2014](#)). Anyway the attentional bias toward the dominant hand should result in faster RT responding with that hand independent of whether the stimuli are positive or negative which could not be confirmed. Additional problem with that attentional explanation comes from the Domino task, where participants paid attention mostly to the gloved hand. This did not result in a main effect of glove and differences in the RTs depending on which hand was gloved.

A third possible explanation comes from the valence hypothesis, according to which emotional valence is processed in different hemispheres. In particular the right prefrontal cortex is associated with negative emotion, whereas the left prefrontal cortex is responsible for the positive emotions ([Reuter-Lorenz & Davidson, 1981](#); [Demaree, Everhart, Youngstrom & Harrison, 2005](#) for review). The idea is that the left hemisphere also controls the right hand, whereas the right hemisphere controls the left hand. The result is congruent with good is right and bad is left valence - response hand mapping, and facilitation of the RTs. It still remains unclear whether the valence hypothesis can be considered separate from the explanation with

the motor fluency because this lateralization can be due to motor fluency or the motor fluency can be an effect of the lateralization.

The aim of the second experiment was to further investigate how stable are the associations between emotional stimuli, and the space around the dominant, and the subordinate hand. Some researcher believed that concepts are stable (e.g., [Barsalou, 1987](#)). But our bodies are not, they are permanently in interaction with the environment. According to the embodied cognition account ([Barsalou, 1999, 2008](#)) concepts are grounded in the sensory-motor experience. This account emphasizes the important role of the interaction between our bodies and the environment for the representation of concepts. So if the human body in the current context is so important than those concepts should be sensitive to changes in the interaction between the body and the environment. Previous research showed that the current experienced motor fluency has a crucial role for the implicit associations between space and valence. When performing a fine motor task for only 12 min. participants showed reversibility of the space-valence associations with right-handers showing “*left is good*” association and left-handers showing “*right is good* associations” ([Casasanto & Chrysikou, 2011](#)). To examine the space-valence associations, an offline forced-choice task consisted of one trial was applied (The Bob task, Experiment 3, [Casasanto, 2009](#)). The aim of the second experiment was to establish whether the induced motor disfluency would influence the RT-profiles of right-handed participants. In order to have better comparison with previous research an offline task was used (The Person task) similar to that used by [Casasanto & Henetz \(2012\)](#), and the same online word categorization task used in experiment 1. In this case the only difference was that the RT categorization task was preceded by a fine-motor task inducing disfluent motor experience of either the dominant hand or the subordinate one. Expectations were that when the glove was on the left hand (subordinate hand disfluency) the space-valence association would be perceived or get even stronger and when the glove was on

the right hand the pattern would be reversed. I found no empirical evidence in this direction. Nevertheless, the interaction between space and valence measured by the RTs disappeared probably as a consequence of the motor disfluency. The RT profiles by right-handed participants looked similar to the “natural” left-handed participants. Unexpectedly when the glove was on the left hand (perceiving natural right-handedness) the effect of the space-valence associations disappeared compared to right-handed participants in experiment 1. The RT profiles were similar but the effects were weaker. Probably this can be due to the peculiarities of the Domino task. That is due to the fact that participants were required to move both hands simultaneously, the ungloved hand experienced also some disfluency, strong enough to cause the effect to disappear. Further research is needed in order to investigate the effect of motor disfluency induced to only one of both hands.

Taken together the results cannot support the idea that changing the fluency reverse the conceptual mapping ([Casasanto & Chrysikou, 2011](#)) but instead the results suggest that it cause space-valence mapping to disappear - both when the glove was on the left hand and when it was on the right hand. Interestingly, the RT profiles for the group who wore the glove on the right hand was similar to those observed by natural left-handers in experiment one. Probably, the experienced motor disfluency is not bounded to the concrete hand but rather than to the whole situation. This leads to the conclusion that the hedonic marking is also not related only to the fluency in the hand motor interaction but is would rather than it is be distributed across the whole situation where disfluency is experienced ([Winkielmann & Cacioppo, 2001](#); [Hayes et al., 2008](#)). Further, the overall RT for people who wore the glove on their right hand were slower than the ones who wore it on the left one which is consistent with the results observed by [Jäncke et al \(1998\)](#) that right handers required more effort when they had to use their non-dominant left-hand. Up to now, in the discussion, I have assumed that the domino task induced motor disfluency. However there is a problem with it as a

method for reversing the hands. No one has tested whether disfluency related to the gloved hand is induced or it spreads to the whole situation. If we assume that it encompasses the whole situation, this can also be an explanation why space-valence mapping disappeared and not reversed.

To conclude, changing the relative fluency influence the conceptual mapping leading to the conclusion that at least the abstract concepts of valence cannot be considered as stable representations, rather they are constructed ad hoc (e.g., [Casasanto & Lupyan, 2015](#)). The authors offered an ad-hoc cognition framework according to which concepts do not exist in the mind, they exist only when we use them. *“Rather than a process of accessing a preformed package of knowledge, instantiating a concept is always a process of activating an ad hoc network of stored information in response to cues in context”* ([Casasanto & Lupyan, 2015, p. 546](#)). There is a debate in the literature whether concepts are stable and context independent (invariantism) or unstable and context dependent (contextualism) (see [Löhr, 2017](#) for review) but according to the currently observed results it can be concluded that the concept of valence is constructed ad-hoc depending on the context. Further research is needed in order to see whether this can be generalized for other abstract concepts.

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Appendix A (Stimuli words)

BLOCK 1: STIMULI WORDS - positive					
Word_BG	Word_EN	frq(Log10)	Word_Leng	Emo_Mean	Ple_Mean
веселие	joy	0,6953	7	6,00	6,25
дарба	gift	0,8355	5	5,58	6,25
добро	good	2,0249	5	5,71	6,21
красота	beauty	1,2711	7	5,46	6,25
късмет	luck	1,5299	6	5,75	6,00
любов	love	1,8669	5	6,75	6,58
мечта	dream	1,2953	5	5,71	6,25
нежност	tenderness	0,7523	7	5,67	6,29
обич	love	1,1121	4	6,25	6,46
очарование	charm	0,4282	10	5,17	5,88
свобода	freedom	1,8272	7	6,21	6,21
смях	laugh	1,5715	4	6,00	6,08
страст	passion	1,1222	6	6,46	5,92
удоволствие	pleasure	1,6846	11	5,88	6,50
хубост	beauty	0,6227	6	5,08	5,75
чудо	miracle	1,5538	4	5,67	5,83
шанс	chance	1,9071	4	5,00	5,88
щастие	happiness	1,5565	6	6,08	6,79
BLOCK 1: STIMULI WORDS - negative					
Word_BG	Word_EN	frq(Log10)	Word_Leng	Emo_Mean	Ple_Mean
гняв	anger	1,2925	4	6,08	2,42
загуба	loss	1,7644	6	6,29	2,46
злоба	malice	0,8587	5	5,50	1,71
криза	crisis	1,9698	5	5,50	1,88
мъка	grief	1,3454	4	6,00	2,00
недоволство	discontent	1,2200	11	5,13	2,92
ненавист	hatred	0,7490	8	5,08	2,08
обида	insult	1,0891	5	5,29	2,33
огорчение	bitterness	0,5820	9	5,67	2,04
омраза	hatred	1,2255	6	6,00	1,50
отчаяние	despair	1,0916	8	5,13	1,71
подлост	mischief	0,3159	7	5,67	1,58
ревност	jealousy	0,8082	7	5,88	2,42
страх	fear	1,8899	5	5,67	2,04
траур	mourning	0,4371	5	5,63	1,75
тревога	anxiety	1,4122	7	5,92	2,17
тъга	sadness	0,9379	4	5,79	2,25
ужас	horror	1,5385	4	5,92	1,75

BLOCK 1: STIMULI WORDS - positive					
Word_BG	Word_EN	frq(Log10)	Word_Leng	Emo_Mean	Ple_Mean
възторг	delight	0,9790	7	5,54	5,83
възхищение	admiration	0,8346	10	5,71	5,75
вяра	faith	1,4914	4	5,88	6,04
грижа	care	1,4101	5	5,50	5,25
дружба	friendship	1,2123	6	5,38	5,96
душа	soul	1,5139	4	5,96	6,00
здраве	health	1,5964	6	5,71	6,63
изненада	surprise	1,7360	8	5,63	5,50
ласка	lassa	0,3386	5	5,75	5,88
милувка	sweetheart	0,0000	7	5,83	6,21
мир	peace	1,8076	3	5,83	6,46
надежда	hope	2,0601	7	5,96	6,08
наслада	delight	0,7822	7	6,08	6,33
радост	joy	1,6599	6	6,08	6,25
усмивка	smile	1,6665	7	6,00	6,42
успех	Good luck	1,9941	5	5,75	6,00
утеха	comfort	0,8399	5	5,25	5,25
хумор	humor	1,2609	5	5,67	6,42
BLOCK 1: STIMULI WORDS - negative					
Word_BG	Word_EN	frq(Log10)	Word_Leng	Emo_Mean	Ple_Mean
беда	trouble	1,0090	4	5,88	2,00
болест	illness	1,4749	6	5,58	1,58
болка	pain	1,6258	5	5,96	1,83
вина	wines	1,7335	4	5,42	2,29
завист	envy	0,9365	6	5,25	1,83
измама	fraud	1,2546	6	5,46	1,79
лъжа	lie	1,2894	4	5,83	2,04
напрежение	tension	1,6883	10	5,04	2,67
нещастие	misfortune	1,0218	8	5,67	2,17
отвращение	disgust	0,9076	10	6,00	1,67
паника	panic	1,2569	6	5,54	2,25
погнуса	disgust	0,4831	7	5,58	1,96
принуда	coercion	0,7721	7	5,25	2,00
скръб	sorrow	0,7761	5	6,00	1,75
смърт	death	1,8484	5	6,42	1,83
срам	shame	1,1923	4	5,79	2,00
страдание	suffering	0,7978	9	5,67	1,58
уплаха	fright	0,5867	6	5,42	2,42

NO STIMULI words					
“WARM UP” words - block1			TRAINING words		
Word_BG	Word_EN	Valence	Word_BG	Word_EN	Valence
извинение	excuse	positive	безработиц		negative
настроение	minded	positive	а	unemployment	
натиск	pressure	negative	вреда	harm	negative
			излет	outing	positive
отегчение	boredom	negative	манипулаци		
спокойствие	calmness	positive	я	manipulation	negative
стрес	stress	negative	полза	advantage	positive
съчувствие	sympathy	positive	предимство	advantage	positive
ярост	fury	negative	разкош	splendor	positive
			угризение	remorse	negative
WARM UP” words - block1					
гадост	dullness	positive			
гнус	abomination	positive			
правда	righteousness	negative			
състрадание	compassion	negative			
творчество	creativity	positive			
уважение	respect	negative			
угризение	remorse	positive			
удар	blow	negative			

Stimuli words: word length measured in number of letters (*Word_Leng*), word frequency (*frq(Log10)*), and emotionality strength (*Ple_Mean*) and valence.

Appendix B (Written instructions)

Word categorization task (Experiment 1 and 2)

На дисплея ще се появяват думи.

Вашата задача е да категоризирате думата като позитивна или негативна и да натискате възможно най-бързо един от следните бутони:

z-3 за позитивна z-3 за негативна

Words will appear on the display.

Your task is to categorize the words as positive and negative and to press as fast as possible one of the followed buttons:

z-3 for positive z-3 for negative

The Domino Task

Домино предизвикателството

Вашата задача е да поставяте домино плочките върху цветните точки пред вас, спазвайки следните правила:

1. Поставяйте доминото върху точките симетрично използвайки двете си ръце едновременно. С лявата ръка поставяйте доминото върху точките в лявата половина на игралното поле, а с дясната върху точките в дясната половина. Опитайте се да движите двете ръце едновременно и в синхрон.
2. Подреждайте плочките ред по ред, започвайки от най-далечния ъгъл на игралното поле и от най-външната към най-вътрешната точка. Плочките трябва да са обърнати с надписите към вас.
3. Ако някоя плочка падне трябва да я поставите отново преди да продължите.
4. Като запълните точките, приберете плочките в кутията и започнете да ги редите отново. Прибирайте точките от дясната половина на игралното поле с дясната ръка и от лявата половина с лявата.
5. Имате 12 мин., за да подредите възможно най-много точки

Изглежда лесно, нали??? НО ето я уловката:

6. През цялото време ще бъдете със скиорска ръкавица на едната ръка (експериментаторът ще ви покаже на коя). Втората ръкавица ще виси закачена за първата. Вашата задача е да поставяте плочките върху точките въпреки трудността предизвикана от ръкавицата.
7. Няколко правила относно ръкавицата:
 - a) Не я сваляйте преди приключване на експеримента
 - b) В никакъв случай не помагайте със свободната си ръка на тази с ръкавицата (не вдигайте плочки със свободната ръка, за да ги поставяте в тази с ръкавицата, не се опитвайте да прихващате висящата ръкавица със свободната ръка).
 - c) Помнете – движете ръцете едновременно и в синхрон до колкото е възможно Ако имате въпроси питайте експериментатора!

Успех!

The Domino Challenge!

In this experiment, your job is to place dominos on the colored dots, standing on their ends, following some *very particular rules!*

1. Please place the dominos on the dots symmetrically, with both hands at the same time. You'll use your left hand for dots on the left side of the boxes,
and your right hand for dots on the right side. Try to keep both hands moving in synchrony.
2. You'll proceed row by row, working from the outermost to the innermost dots, starting from the corners farthest away from you. Be sure the dominos are facing you.
3. If a domino falls over (or falls on the floor!) you must stand it up again on the correct dot before continuing on. Be sure to use the correct hand!
4. Once you've filled the whole board, please put all of the dominos into the box again, and start over! Use your left hand to put dominos from the left side back into the boxes, and your right hand for dominos on the right side.
5. You'll have 12 minutes to fill up as many rows of dominos as possible.

Sounds easy, yes? But here's the catch!!

6. The whole time, you'll be wearing a ski glove on one of your hands (the experimenter will show you which one). There's another ski glove attached to the one you'll be wearing, which will be dangling from your wrist. Your job is to meet the Domino Challenge in spite of this handicap!
7. Some rules about the glove:
 - a. Do not take the glove off until the experiment is over.
 - b. Do not use your free hand to help your gloved hand in any way (e.g., don't pick up dominos with your free hand and place them in your gloved hand, don't use your free hand to control the dangling glove...)
 - c. Remember to keep your two hands in synchrony as much as you can!

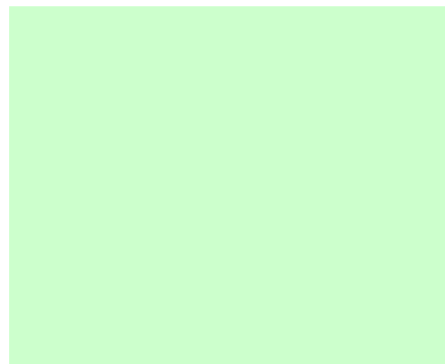
If you have any questions about these instructions, please ask the experimenter.

Good luck

The Person Task – original instruction

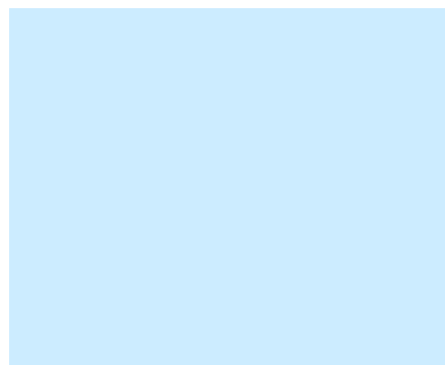
Представете си човек, който много цените, уважавате и обичате. Човек, с който сте споделяли хубави мигове, на когото държите и когото се радвате, че познавате.

Напишете първата буква от името му на ЗЕЛЕНОТО листче и поставете листчето в една от кутиите пред вас.



Сега си намислете някой, който не харесвате. Човек, който смятате за неприятен, когото не цените. Човек, който ви кара да се чувствате зле и с когото не бихте искали да общувате.

Напишете първата буква от името му на СИНЬОТО листче и поставете листчето в една от кутиите пред вас.



The Person Task – english instruction

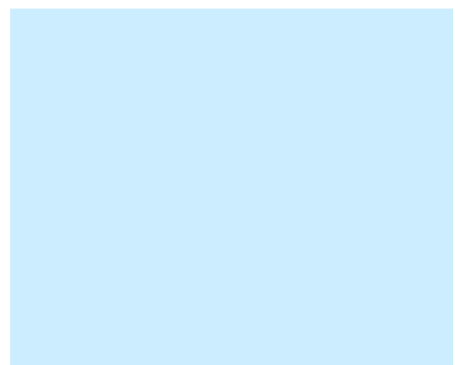
Imagine a person who you highly, appreciate respect and love. The person with whom you shared good moments, whom you hold and whom you are happy to know.

Write the first letter of his/hers name on the GREEN sheet and put the sheet in one of the boxes in front of you.



Now think of someone you do not like. The person you think is unpleasant, whom he does not appreciate. A person who makes you feel bad and with whom you do not want to communicate.

Write the first letter of his/hers name on the SYNTHÉ list and place the sheet in one of the boxes in front of you.



Appendix C (NBU Handedness Inventory)

ИЛ №: _____

NBUHI-2
Дата: __ | __ | 20__

Въпросник

Първа част.

Моля, прочетете всеки от въпросите по-долу. Посочете своите предпочитания към използване на съответните ръка, крак и т.н., при извършване на всяка една от описаните дейности, като поставите знак (✓) в съответната колонка. Ако не сте сигурни с коя ръка бихте извършили някоя от дейностите, опитайте се да я извършите (имитирате). Ако имате някакви въпроси, моля обърнете се към експериментатора.

№	Въпрос	лявата	и двете	дясната
1	С коя ръка пишете четливо?			
2	С коя ръка хвърляте топка, за да улучите някаква цел?			
3	С коя ръка държите тенис-ракета или хилка за федербал?			
4	С коя ръка държите клечка кибрит, когато я запалвате?			
5	С коя ръка държите ножици?			
6	Коя ръка движите, когато вдявате конец (независимо дали в нея държите иглата или конца)?			
7	Ако държите метла с дълга дръжка, коя ръка ще е отгоре?			
8	Коя ръката е отгоре, когато държите лопата, за да пренесете пясък?			
9	С коя ръка раздавате карти?			
10	С коя ръка държите чука, когато забивате пирон?			
11	Когато работите с компютър, с коя ръка движите мишката?			
12	С коя ръка държите четката за зъби?			
13	С коя ръка развивате капачката на			

	буркан?			
14	С коя ръка държите лъжица?			
15	С кой крак ритате топка, за да уцелите нещо?			
16	Ако ви се налага да се качите на стол, кой ще е първия крак, който ще качите?			
17	С кое око гледате през телескоп?			
18	Ако искате да погледнете през гърлото на бутилка, за да видите дали не е празна, с кое око бихте погледнали?			
19	В кое ухо ще поставите слушалка на смартфон или mp3 плейър (ако напр. с някого делите две слушалки)?			
20	Ако искате да чуете как бие сърцето на някого, кое ухо ще поставите на гърдите му?			

Втора част

Моля отговорете на допълнителните въпроси по-долу.

1. Считате ли себе си за:

..... деснорък
..... леворък
..... еднакво добре използващ двете си ръце

2. Някога променяли ли сте своята ръкост? Ако „Да”, кога и защо?

.....
.....

3. Сещате ли се за някоя дейност, която да не е включена в този списък, за която Вие редовно използвате *не* предпочитаната от Вас ръка?

.....
.....

Благодарим ви за участието!

Subject ID: _____
___ | 200_

Date: __ |

Questionnaire

Part one.

Simply read each of the questions below. Decide which hand, foot, etc. you use for each activity and then put a check mark (✓) next to the answer that describes you the best. If you are unsure of any answer, try to act out the action. If you have some questions please refer to the experimenter.

No	Question	Left	Both	Right
1	Which hand do you use to write a letter legibly?			
2	Which hand do you use to throw a ball to hit a target?			
3	Which hand do you use to hold a racket in tennis squash or badminton?			
4	Which hand do you use to hold a match whilst striking it?			
5	Which hand do you use to cut with scissors?			
6	Which hand do you use to guide a thread through the eye of a needle (or guide needle on to thread)?			
7	Which hand do you use at the top of a broom while sweeping?			
8	Which hand do you use at the top of the shovel while moving sand?			
9	Which hand do you use to deal playing cards?			
10	Which hand do you use to hammer a nail into wood?			
11	Which hand do you use to hold a toothbrush while cleaning your teeth?			
12	Which hand do you use to unscrew the lid of a jar?			
13	With which hand would you use a spoon?			

14	With which foot would you kick a ball to hit a target?			
15	If you had to step up onto a chair, which foot would you place on the chair first?			
16	Which eye would you use to look through a telescope?			
17	If you had to look into a dark bottle to see how full it was, which eye would you use?			
18	If you wanted to listen in on a conversation going on behind a closed door, which ear would you place against the door?			
19	Into which ear would you place the earphone of a radio or a CD player?			
20	If you wanted to hear someone's heartbeat which ear would you place against their chest?			

Part two

Please answer to the following additional questions.

Are you:

..... right-hander
..... left-hander
..... ambidextrous

2. Have you ever changed your handedness. If yes, when and why?

2. If you use your *left* hand or foot, etc. for *all* of the actions above, are there any one-handed actions for which you use your *right* hand or foot, etc.? Please record them here.

Thank you for your cooperation!

Appendix D (Linear Mixed Model)

The linear mixed models analysis was conducted using the lme4 package (Bates & Maechler, 2010).

1) Experiment 1

There were 3 fixed effects: valence (positive vs. negative), response hand (left vs. right), and handedness (left, right vs. ambidextrous), and two random effects: participants and items.

Final R code for models:

Model H_0 = `lmer(Stim.RT~ Response Hand + Valence + Handedness + Response Hand * Valence + Response Hand * Handedness + Handedness * Valence + (1|Subject)+(1|Stimuli))`

Model H_A = `lmer(Stim.RT~ Response Hand + Valence + Handedness + Response Hand * Valence * Handedness + (1|Subject)+(1|Stimuli))`.

The Model H_A includes all effects and interactions; Model H_0 is the same but without the three-way interaction. The results showed that the model with the three-way interaction can better account for the data.

$$[\chi^2(1, N = 91) = 30.643, p = .000]$$

2) Experiment 2

There were 3 fixed effects: valence (positive vs. negative), response hand (left vs. right), and glove (left vs. right) and two random effects: participants and items.

Final R code for models:

Model H_0 = `lmer(Stim.RT~ Response Hand+ Valence +Glove + Response Hand * Valence + Response Hand * Handedness + Handedness * Valence +(1|Subject)+(1|Stimuli))`

Model H_A = `lmer(Stim.RT~ Response Hand + Valence + Glove + Response Hand * Valence`

* $Glove + (1|Subject) + (1|Stimuli)$).

The Model H_A includes all effects and interactions; Model H_0 is the same but without the three-way interaction. The results showed weaker effect in favor of Model H_A [$\chi^2(1, N = 71) = 4.3388, p = .037$].